DERIVED COMPOSITION ISOMORPHISM RG_! COMPOSED WITH RF_! EQUALS R(GF)_! FOR SEPARATED FINITE TYPE MORPHISMS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-etale.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Nagata compactification (More on Flatness, Theorem flat-theorem-nagata, canonical tag 0F41). cited at proof lines 3385 and 3446-3448. Provides the compactifications of Y over Z and X over bar-Y, including the dominating common choice.
- Well-definedness of the compactification construction (Lemma lemma-shriek-well-defined, canonical tag 0F7I). cited at proof line 3538. Fixes how the five square isomorphisms enter the construction being compared.

Source reference. Original lemma 04VJ: sources/stacks-chow/tags.txt, source lines 6327–6327. Cited at proof line 3451: equality criterion used when choosing the dominating compactification.

Source reference. Original lemma 04VI: sources/stacks-chow/tags.txt, source lines 6326–6326. Cited at proof line 3450: localized-category morphisms used in the dominating-compactification choice.

Source reference. Original lemma 0ATU: sources/stacks-chow/tags.txt, source lines 13627–13627. Cited at proof line 3448: the common compactification of Y over Z dominating both choices.

Source reference. Original lemma 0A9Z: sources/stacks-chow/tags.txt, source lines 12967–12967. Cited at proof line 3386: compactifyability used twice to compactify X over bar-Y.

Source reference. Original lemma 0ATV: sources/stacks-chow/tags.txt, source lines 13628–13628. Cited at proof line 3449: the multiplicative-system property used in the dominating choice.

Source reference. Original theorem 0F41: sources/stacks-chow/tags.txt, source lines 18886–18886. Cited at proof line 3385: existence of the compactifications.

Source reference. Original lemma 0F50: sources/stacks-chow/tags.txt, source lines 18920–18920. Cited at proof line 3400: the second equality of the four-step chain.

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Source reference. Original lemma 0F7G: sources/stacks-chow/tags.txt, source lines 19006–19006. Cited at proof lines 3527-3528: the same map as lemma-shriek-proper-and-open when the left vertical arrow is the identity.

Source reference. Original lemma 0F7B: sources/stacks-chow/tags.txt, source lines 19001–19001. Cited at proof lines 3402 and 3526: the third equality of the chain and the square isomorphisms gamma.

Source reference. Original lemma 0F7C: sources/stacks-chow/tags.txt, source lines 19002–19002. Cited at proof line 3556: composition rule used in the pentagon collapse.

Source reference. Original lemma 0F7D: sources/stacks-chow/tags.txt, source lines 19003–19003. Cited at proof line 3557: composition rule used in the pentagon collapse.

Source reference. Original lemma 0F7I: sources/stacks-chow/tags.txt, source lines 19008–19008. Cited at proof line 3538: how the construction consumes the square isomorphisms.

Source reference. Original remark 0F7E: sources/stacks-chow/tags.txt, source lines 19004–19004. Cited at proof lines 3559 and 3578: notation conventions and the single-intersection-point argument.

Lemma 0.2. *Let $f : X \to Y$ and $g : Y \to Z$ be separated morphisms of finite type of quasi-compact and quasi-separated schemes. Then there is a canonical isomorphism $Rg_! \circ Rf_! \to R(g \circ f)_!$.*

Proof. Choose a compactification $i : Y \to \overline{Y}$ of Y over Z. Choose a compactification $X \to \overline{X}$ of X over $\overline{Y}$. This uses More on Flatness, Theorem 0F41 and Lemma 0A9Z twice. Let U be the inverse image of Y in $\overline{X}$ so that we get the commutative diagram

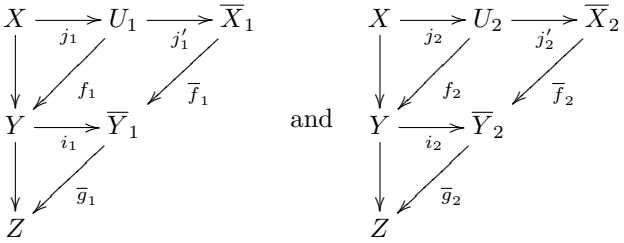
$$\begin{array}{ccccc} X & \xrightarrow{j} & U & \xrightarrow{j'} & \overline{X} \\ {\scriptstyle f}\downarrow & \swarrow{\scriptstyle f'} & & \swarrow{\scriptstyle \overline{f}} & \\ Y & & \xrightarrow{i} & & \overline{Y} \\ {\scriptstyle g}\downarrow & & \swarrow{\scriptstyle \overline{g}} & & \\ Z & & & & \end{array}$$

Then we have

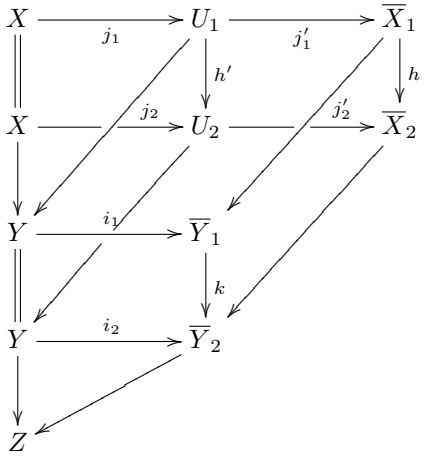
$$\begin{aligned} R(g \circ f)_! &= R(\overline{g} \circ \overline{f})_* \circ (j' \circ j)_! \\ &= R\overline{g}_* \circ R\overline{f}_* \circ j'_! \circ j_! \\ &= R\overline{g}_* \circ i_! \circ Rf'_* \circ j_! \\ &= Rg_! \circ Rf_! \end{aligned}$$

The first equality is the definition of $R(g \circ f)_!$. The second equality uses the identifications $R(\overline{g} \circ \overline{f})_* = R\overline{g}_* \circ R\overline{f}_*$ and $(j' \circ j)_! = j'_! \circ j_!$ of Lemma 0F50. The identification $i_! \circ Rf'_* \to R\overline{f}_* \circ j_!$ used in the third equality is Lemma 0F7B. The final fourth equality is the definition of $Rg_!$ and $Rf_!$. To finish the proof we show that this isomorphism is independent of choices made.

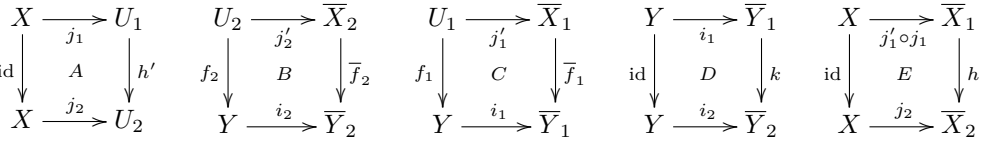
Suppose we have two diagrams



We can first choose a compactification $i : Y \to \overline{Y}$ of Y over Z which dominates both $\overline{Y}_1$ and $\overline{Y}_2$, see More on Flatness, Lemma 0ATU. By More on Flatness, Lemma 0ATV and Categories, Lemmas 04VI and 04VJ we can choose a compactification $X \to \overline{X}$ of X over $\overline{Y}$ with morphisms $\overline{X} \to \overline{X}_1$ and $\overline{X} \to \overline{X}_2$ and such that the composition $\overline{X} \to \overline{Y} \to \overline{Y}_1$ is equal to the composition $\overline{X} \to \overline{X}_1 \to \overline{Y}_1$ and such that the composition $\overline{X} \to \overline{Y} \to \overline{Y}_2$ is equal to the composition $\overline{X} \to \overline{X}_2 \to \overline{Y}_2$. Thus we see that it suffices to compare the maps determined by our diagrams when we have a commutative diagram as follows



Each of the squares



gives rise to an isomorphism as follows

$$\gamma_A : j_{2,!} \to Rh'_* \circ j_{1,!}$$
$$\gamma_B : i_{2,!} \circ Rf_{2,*} \to R\overline{f}_{2,*} \circ j'_{2,!}$$
$$\gamma_C : i_{1,!} \circ Rf_{1,*} \to R\overline{f}_{1,*} \circ j'_{1,!}$$
$$\gamma_D : i_{2,!} \to Rk_* \circ i_{1,!}$$
$$\gamma_E : j_{2,!} \to Rh_* \circ (j'_1 \circ j_1)_!$$

by applying the map from Lemma 0F7B (which is the same as the map in Lemma 0F7G in case the left vertical arrow is the identity). Let us write

$$\begin{aligned} F_1 &= Rf_{1,*} \circ j_{1,!} \\ F_2 &= Rf_{2,*} \circ j_{2,!} \\ G_1 &= R\overline{g}_{1,*} \circ i_{1,!} \\ G_2 &= R\overline{g}_{2,*} \circ i_{2,!} \\ C_1 &= R(\overline{g}_1 \circ \overline{f}_1)_* \circ (j'_1 \circ j_1)_! \\ C_2 &= R(\overline{g}_2 \circ \overline{f}_2)_* \circ (j'_2 \circ j_2)_! \end{aligned}$$

The construction given in the first paragraph of the proof and in Lemma 0F7I uses

(1) γ_C for the map $G_1 \circ F_1 \to C_1$,
(2) γ_B for the map $G_2 \circ F_2 \to C_2$,
(3) γ_A for the map $F_2 \to F_1$,
(4) γ_D for the map $G_2 \to G_1$, and
(5) γ_E for the map $C_2 \to C_1$.

This implies that we have to show that the diagram

$$\begin{array}{ccc} C_2 & \xrightarrow{\gamma_E} & C_1 \\ {\scriptstyle \gamma_B}\uparrow & & \uparrow{\scriptstyle \gamma_C} \\ G_2 \circ F_2 & \xrightarrow{\gamma_D \circ \gamma_A} & G_1 \circ F_1 \end{array}$$

is commutative. We will use Lemmas 0F7C and 0F7D and with (abuse of) notation as in Remark 0F7E (in particular dropping $\star$ products with identity transformations from the notation). We can write $\gamma_E = \gamma_F \circ \gamma_A$ where

$$\begin{array}{ccc} U_1 & \xrightarrow{j'_1} & \overline{X}_1 \\ {\scriptstyle h'}\downarrow & F & \downarrow{\scriptstyle h} \\ U_2 & \xrightarrow{j'_2} & \overline{X}_2 \end{array}$$

Thus we see that

$$\gamma_E \circ \gamma_B = \gamma_F \circ \gamma_A \circ \gamma_B = \gamma_F \circ \gamma_B \circ \gamma_A$$

the last equality because the two squares A and B only intersect in one point (similar to the last argument in Remark 0F7E). Thus it suffices to prove that $\gamma_C \circ \gamma_D = \gamma_F \circ \gamma_B$. Since both of these are equal to the map for the square

$$\begin{array}{ccc} U_1 & \longrightarrow & \overline{X}_1 \\ \downarrow & & \downarrow \\ Y & \longrightarrow & \overline{Y}_2 \end{array}$$

we conclude. □